—1—

Miscellaneous mathematical features

1.1 Matrices

The amsmath package provides some environments for matrices beyond the basic `array` environment of LaTeX. The `pmatrix`, `bmatrix`, `Bmatrix`, `vmatrix` and `Vmatrix` have (respectively) ( ), [ ], { }, | |, and ‖ ‖ delimiters built in. For naming consistency there is a `matrix` environment sans delimiters. This is not entirely redundant with the `array` environment; the matrix environments all use more economical horizontal spacing than the rather prodigal spacing of the `array` environment. Also, unlike the `array` environment, you don't have to give column specifications for any of the matrix environments; by default you can have up to 10 centered columns.[1] (If you need left or right alignment in a column or other special formats you may use `array`, or the `mathtools` package which provides `*` variants of these environments with an optional argument to specify left or right alignment.)

To produce a small matrix suitable for use in text, there is a `smallmatrix` environment (e.g., $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$) that comes closer to fitting within a single text line than a normal matrix. Delimiters must be provided. (The `mathtools` package provides `p`,`b`,`B`,`v`,`V` versions of `smallmatrix`, as well as `*` variants as described above.) The above example was produced by

```
\bigl( \begin{smallmatrix}
  a&b\\ c&d
\end{smallmatrix} \bigr)
```

`\hdotsfor{`⟨*number*⟩`}` produces a row of dots in a matrix spanning the given number of columns. For example,

$$\begin{matrix} a & b & c & d \\ e & \cdot & \cdot & \cdot \end{matrix}$$

```
\begin{matrix} a&b&c&d\\
e&\hdotsfor{3} \end{matrix}
```

The spacing of the dots can be varied through use of a square-bracket option, for example, `\hdotsfor[1.5]{3}`. The number in square brackets will be used as a multiplier (i.e., the normal value is 1.0).

$$\begin{pmatrix} D_1t & -a_{12}t_2 & \dots & -a_{1n}t_n \\ -a_{21}t_1 & D_2t & \dots & -a_{2n}t_n \\ \dots & \dots & \dots & \dots \\ -a_{n1}t_1 & -a_{n2}t_2 & \dots & D_nt \end{pmatrix}, \tag{1.1}$$

```
\begin{pmatrix} D_1t&-a_{12}t_2&\dots&-a_{1n}t_n\\
-a_{21}t_1&D_2t&\dots&-a_{2n}t_n\\
\hdotsfor[2]{4}\\
-a_{n1}t_1&-a_{n2}t_2&\dots&D_nt\end{pmatrix}
```

[1] More precisely: The maximum number of columns in a matrix is determined by the counter `MaxMatrixCols` (normal value = 10), which you can change if necessary using LaTeX's `\setcounter` or `\addtocounter` commands.

1.2 Math spacing commands

The amsmath package slightly extends the set of math spacing commands, as shown below. Both the spelled-out and abbreviated forms of these commands are robust, and they can also be used outside of math.

Abbrev.	Spelled out	Example	Abbrev.	Spelled out	Example
	no space	$\Rightarrow\Leftarrow$		no space	$\Rightarrow\Leftarrow$
`\,`	`\thinspace`	$\Rightarrow\!\Leftarrow$	`\!`	`\negthinspace`	$\Rightarrow\!\!\Leftarrow$
`\:`	`\medspace`	$\Rightarrow\!\Leftarrow$		`\negmedspace`	$\Rightarrow\!\!\Leftarrow$
`\;`	`\thickspace`	$\Rightarrow\!\Leftarrow$		`\negthickspace`	$\Rightarrow\!\!\Leftarrow$
	`\quad`	$\Rightarrow\quad\Leftarrow$			
	`\qquad`	$\Rightarrow\qquad\Leftarrow$			

For the greatest possible control over math spacing, use `\mspace` and 'math units'. One math unit, or `mu`, is equal to 1/18 em. Thus to get a negative `\quad` you could write `\mspace{-18.0mu}`.

1.3 Dots

For preferred placement of ellipsis dots (raised or on-line) in various contexts there is no general consensus. It may therefore be considered a matter of taste. By using the semantically oriented commands

- `\dotsc` for "dots with commas"
- `\dotsb` for "dots with binary operators/relations"
- `\dotsm` for "multiplication dots"
- `\dotsi` for "dots with integrals"
- `\dotso` for "other dots" (none of the above)

instead of `\ldots` and `\cdots`, you make it possible for your document to be adapted to different conventions on the fly, in case (for example) you have to submit it to a publisher who insists on following house tradition in this respect. The default treatment for the various kinds follows American Mathematical Society conventions:

```
Then we have the series $A_1, A_2,
\dotsc$, the regional sum $A_1
+A_2 +\dotsb $, the orthogonal
product $A_1 A_2 \dotsm $, and
the infinite integral
\[\int_{A_1}\int_{A_2}\dotsi\].
```

Then we have the series $A_1, A_2, \dotsc$, the regional sum $A_1 + A_2 + \dotsb$, the orthogonal product $A_1 A_2 \dotsm$, and the infinite integral

$$\int_{A_1}\int_{A_2}\dotsi.$$

For most situations, the undifferentiated `\dots` can be used, and amsmath will output the most suitable form based on the immediate context; if an inappropriate form results, it can be corrected after examining the output.

1.4 Nonbreaking dashes

A command `\nobreakdash` is provided to suppress the possibility of a line break after the following hyphen or dash. For example, if you write 'pages 1–9' as `pages 1\nobreakdash--9` then a line break will never occur between the dash and the 9. You can also use `\nobreakdash` to prevent undesirable hyphenations in combinations like `$p$-adic`. For frequent use, it's advisable to make abbreviations, e.g.,

```
\newcommand{\p}{$p$\nobreakdash}% for "\p-adic"
\newcommand{\Ndash}{\nobreakdash--}% for "pages 1\Ndash 9"
%    For "\n dimensional" ("n-dimensional"):
\newcommand{\n}[1]{$n$\nobreakdash-\hspace{0pt}}
```

The last example shows how to prohibit a line break after the hyphen but allow normal hyphenation in the following word. (It suffices to add a zero-width space after the hyphen.)

1.5 Accents in math

In ordinary LaTeX the placement of the second accent in doubled math accents is often poor. With the amsmath package you will get improved placement of the second accent: $\hat{\hat{A}}$ (`\hat{\hat{A}}`).

The commands `\dddot` and `\ddddot` are available to produce triple and quadruple dot accents in addition to the `\dot` and `\ddot` accents already available in LaTeX.

To get a superscripted hat or tilde character, load the `amsxtra` package and use `\sphat` or `\sptilde`. Usage is `A\sphat` (note the absence of the `^` character).

To place an arbitrary symbol in math accent position, or to get under accents, see the `accents` package by Javier Bezos. (`amsmath` must be loaded before `accents`.)

1.6 Roots

In ordinary LaTeX the placement of root indices is sometimes not so good: $\sqrt[\beta]{k}$ (`\sqrt[\beta]{k}`). In the amsmath package `\leftroot` and `\uproot` allow you to adjust the position of the root:

```
\sqrt[\leftroot{-2}\uproot{2}\beta]{k}
```

will move the beta up and to the right: $\sqrt[\beta]{k}$. The negative argument used with `\leftroot` moves the β to the right. The units are a small amount that is a useful size for such adjustments.

1.7 Boxed formulas

The command `\boxed` puts a box around its argument, like `\fbox` except that the contents are in math mode:

$$\boxed{\eta \leq C(\delta(\eta) + \Lambda_M(0, \delta))} \tag{1.2}$$

```
\boxed{\eta \leq C(\delta(\eta) +\Lambda_M(0,\delta))}
```

1.8 Over and under arrows

Basic LaTeX provides `\overrightarrow` and `\overleftarrow` commands. Some additional over and under arrow commands are provided by the amsmath package to extend the set:

```
\overleftarrow          \underleftarrow
\overrightarrow         \underrightarrow
\overleftrightarrow     \underleftrightarrow
```

1.9 Extensible arrows

`\xleftarrow` and `\xrightarrow` produce arrows that extend automatically to accommodate unusually wide subscripts or superscripts. These commands take one optional argument (the subscript) and one mandatory argument (the superscript, possibly empty):

$$A \xleftarrow{n+\mu-1} B \xrightarrow[T]{n\pm i-1} C \tag{1.3}$$

```
\xleftarrow{n+\mu-1}\quad \xrightarrow[T]{n\pm i-1}
```

1.10 Affixing symbols to other symbols

LaTeX provides `\stackrel` for placing a superscript above a binary relation. In the amsmath package there are somewhat more general commands, `\overset` and `\underset`, that can be used to place one symbol above or below another symbol, whether it's a relation or something else. The input `\overset{*}{X}` will place a superscript-size $*$ above the X: $\overset{*}{X}$; `\underset` is the analog for adding a symbol underneath. The command `\overunderset` is a combination of these, taking three arguments to place superscript sized expressions above and below the same base.

See also the description of `\sideset` in §**??**.

1.11 Fractions and related constructions

1.11.1 The `\frac`, `\dfrac`, and `\tfrac` commands

The `\frac` command, which is in the basic command set of LaTeX, takes two arguments—numerator and denominator—and typesets them in normal fraction form. The amsmath package provides also `\dfrac` and `\tfrac` as convenient abbreviations for `{\displaystyle\frac ... }` and `{\textstyle\frac ... }`.

$$\frac{1}{k}\log_2 c(f) \quad \tfrac{1}{k}\log_2 c(f) \quad \sqrt{\frac{1}{k}\log_2 c(f)} \quad \sqrt{\dfrac{1}{k}\log_2 c(f)} \tag{1.4}$$

```
\begin{equation}
\frac{1}{k}\log_2 c(f)\;\tfrac{1}{k}\log_2 c(f)\;
\sqrt{\frac{1}{k}\log_2 c(f)}\;\sqrt{\dfrac{1}{k}\log_2 c(f)}
\end{equation}
```

1.11.2 The \binom, \dbinom, and \tbinom commands

For binomial expressions such as $\binom{n}{k}$ amsmath has \binom, \dbinom and \tbinom:

$$2^k - \binom{k}{1}2^{k-1} + \binom{k}{2}2^{k-2} \tag{1.5}$$

```
2^k-\binom{k}{1}2^{k-1}+\binom{k}{2}2^{k-2}
```

1.11.3 The \genfrac command

The capabilities of \frac, \binom, and their variants are subsumed by a generalized fraction command \genfrac with six arguments. The last two correspond to \frac's numerator and denominator; the first two are optional delimiters (as seen in \binom); the third is a line thickness override (\binom uses this to set the fraction line thickness to 0—i.e., invisible); and the fourth argument is a mathstyle override: integer values 0–3 select respectively \displaystyle, \textstyle, \scriptstyle, and \scriptscriptstyle. If the third argument is left empty, the line thickness defaults to 'normal'.

> \genfrac{*left-delim*}{*right-delim*}{*thickness*}{*mathstyle*}
> {*numerator*}{*denominator*}

To illustrate, here is how \frac, \tfrac, and \binom might be defined.

```
\newcommand{\frac}[2]{\genfrac{}{}{}{}{#1}{#2}}
\newcommand{\tfrac}[2]{\genfrac{}{}{}{1}{#1}{#2}}
\newcommand{\binom}[2]{\genfrac{(}{)}{0pt}{}{#1}{#2}}
```

If you find yourself repeatedly using \genfrac throughout a document for a particular notation, you will do yourself a favor (and your publisher) if you define a meaningfully-named abbreviation for that notation, along the lines of \frac and \binom.

The primitive generalized fraction commands \over, \overwithdelims, \atop, \atopwithdelims, \above, \abovewithdelims produce warning messages if used with the amsmath package, for reasons discussed in technote.tex.